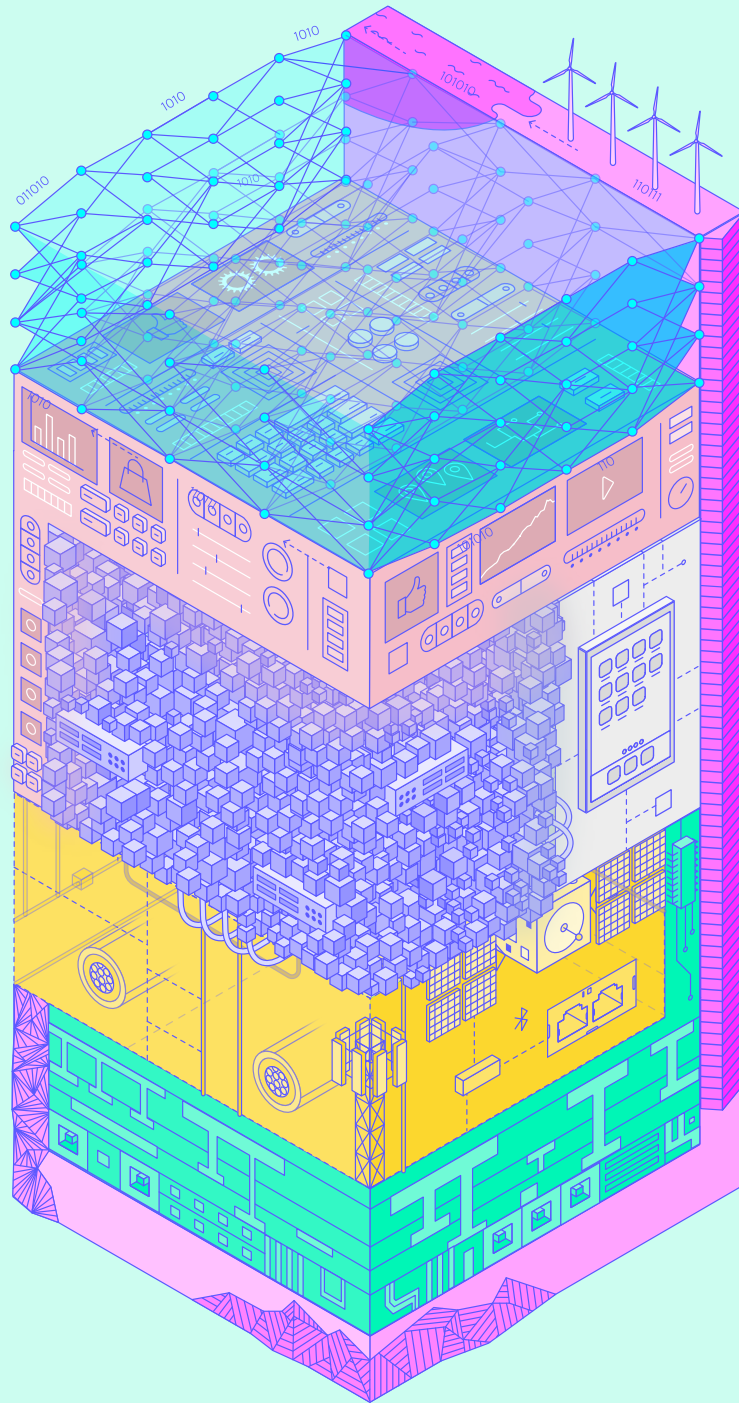


EuroStack – A European Alternative for Digital Sovereignty



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Commissioned by

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Executive summary

The EuroStack initiative presents a bold vision for Europe's digital future, aiming to establish the continent as a leader in digital sovereignty. This comprehensive strategy seeks to foster innovation, strengthen strategic autonomy, and build inclusive partnerships to overcome Europe's reliance on external technologies and position itself at the forefront of the global digital economy.

Currently, over 80% of Europe's digital infrastructure and technologies are imported, creating systemic vulnerabilities and hampering the region's capacity for innovation and self-reliance. The EuroStack initiative directly addresses these challenges by providing a comprehensive strategy to strengthen Europe's competitiveness, secure essential resources, and build a resilient and forward-looking digital ecosystem.

At the core of the EuroStack blueprint is a digitally sovereign Europe built on interconnected layers of advanced technologies, ranging from semiconductors and artificial intelligence (AI) to cloud computing and quantum systems. This approach prioritizes sustainability, inclusivity, and interoperability, ensuring Europe's digital future aligns with its democratic values, social equity goals, and economic aspirations. Crucially, the strategy recognizes that full self-sufficiency is neither feasible nor desirable in a globalized world. Instead, the EuroStack initiative promotes building strategic capabilities while maintaining beneficial international collaborations.

Artificial intelligence stands out as the cornerstone of the EuroStack vision, offering unparalleled opportunities to revolutionize Europe's digital ecosystem. AI's transformative potential extends across all sectors, from manufacturing and healthcare to public services and energy management. By investing in sovereign AI ecosystems, Europe can unlock efficiencies, create smarter public services, and promote indigenous innovation that aligns with European values of transparency, accountability, and privacy.

With 70% of foundational AI models developed in the United States, and China rapidly expanding its share, Europe must focus on fostering homegrown AI capabilities. AI is a strategic enabler that powers data-driven decision-making, optimizes industrial processes, and accelerates research and development. By creating sovereign AI platforms and federated data spaces, the EuroStack initiative aims to reduce dependencies on foreign providers, protect intellectual property, and position Europe as a leader in public interest AI.

Moreover, AI has the potential to significantly narrow the innovation gap between Europe and its global competitors. As highlighted in the Mario Draghi Report of 2024, Europe's slower productivity growth relative to the United States stems from underinvestment in cutting-edge technology and limited translation of Research & Development (R&D) into market successes. AI offers the means to reverse this trend, providing the tools needed to accelerate

innovation and bridge this gap. By integrating AI with Europe's industrial base, including small and medium-sized enterprises, the EuroStack initiative lowers technology adoption costs, enhances productivity, and generates valuable knowledge spillovers that drive economic growth.

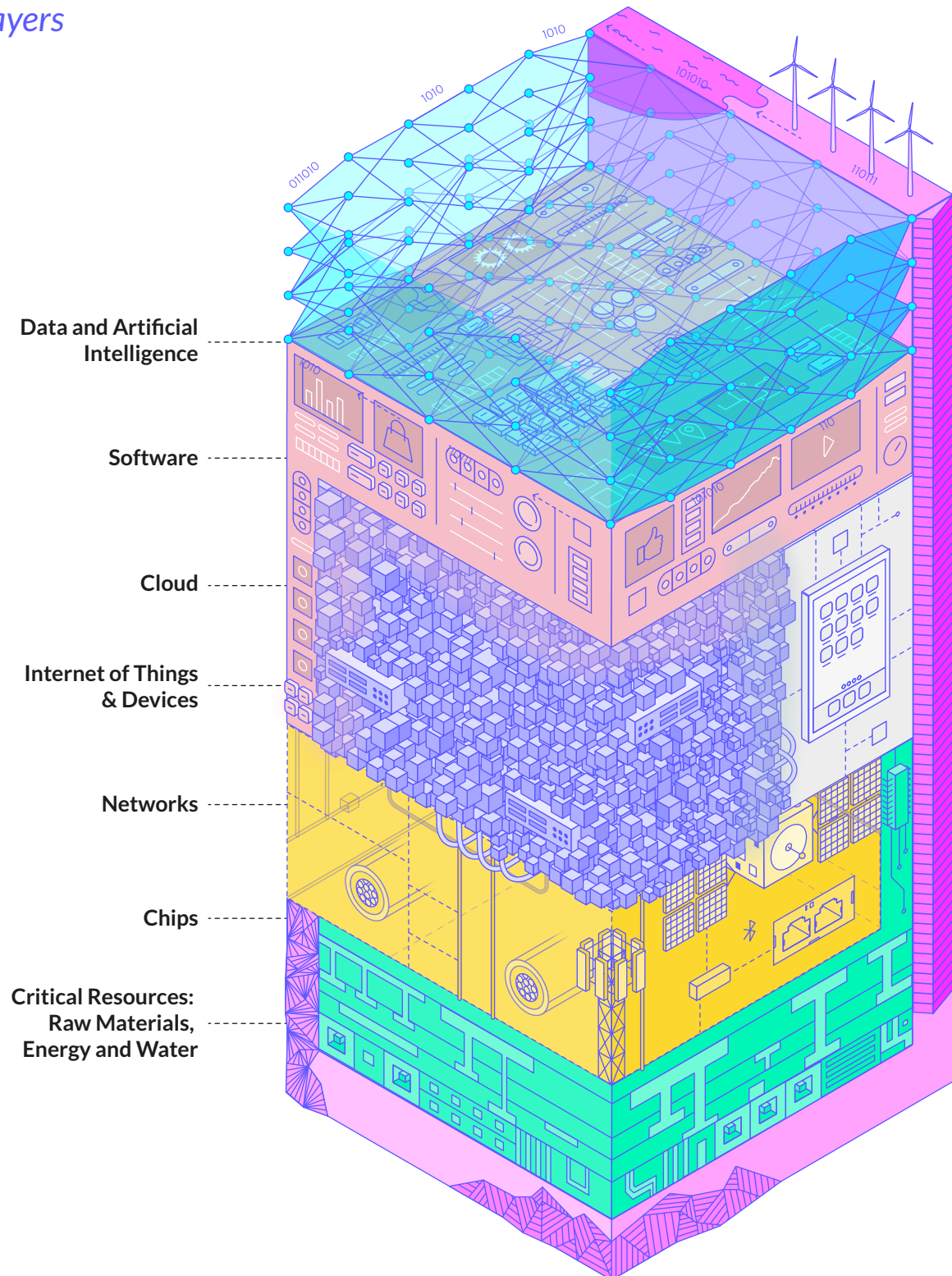
With a proposed investment of €300 billion over the next decade, EuroStack aspires to deliver transformative economic, social, and environmental benefits. These include generating high-skilled jobs, reducing dependencies, and advancing critical sectors such as energy, manufacturing, and healthcare. The initiative also aims to position Europe as a leader in values-driven, citizen-focused innovation, shaping a digital future that prioritizes privacy, trust, and accountability.

The EuroStack model champions a more balanced power dynamic by enabling smaller firms to access advanced tools and technologies, creating a level playing field that fosters entrepreneurship and innovation. By encouraging knowledge-sharing and inclusivity, the initiative aims to transform Europe's digital economy into an ecosystem that creates long-term value rather than simply extracting it.

The EuroStack initiative is not just a policy framework; it is a call to action for a united Europe to take control of its digital destiny. By prioritizing strategic autonomy and integrating emerging technologies such as AI, quantum computing, and advanced cloud solutions, the EuroStack strategy ensures Europe is prepared to confront future challenges while capitalizing on emerging opportunities. This is Europe's moment to create a digital future that embodies its democratic ideals, economic aspirations, and environmental commitments, thereby ultimately strengthening its sovereignty and shaping its role in a multipolar world.

The current digital stack

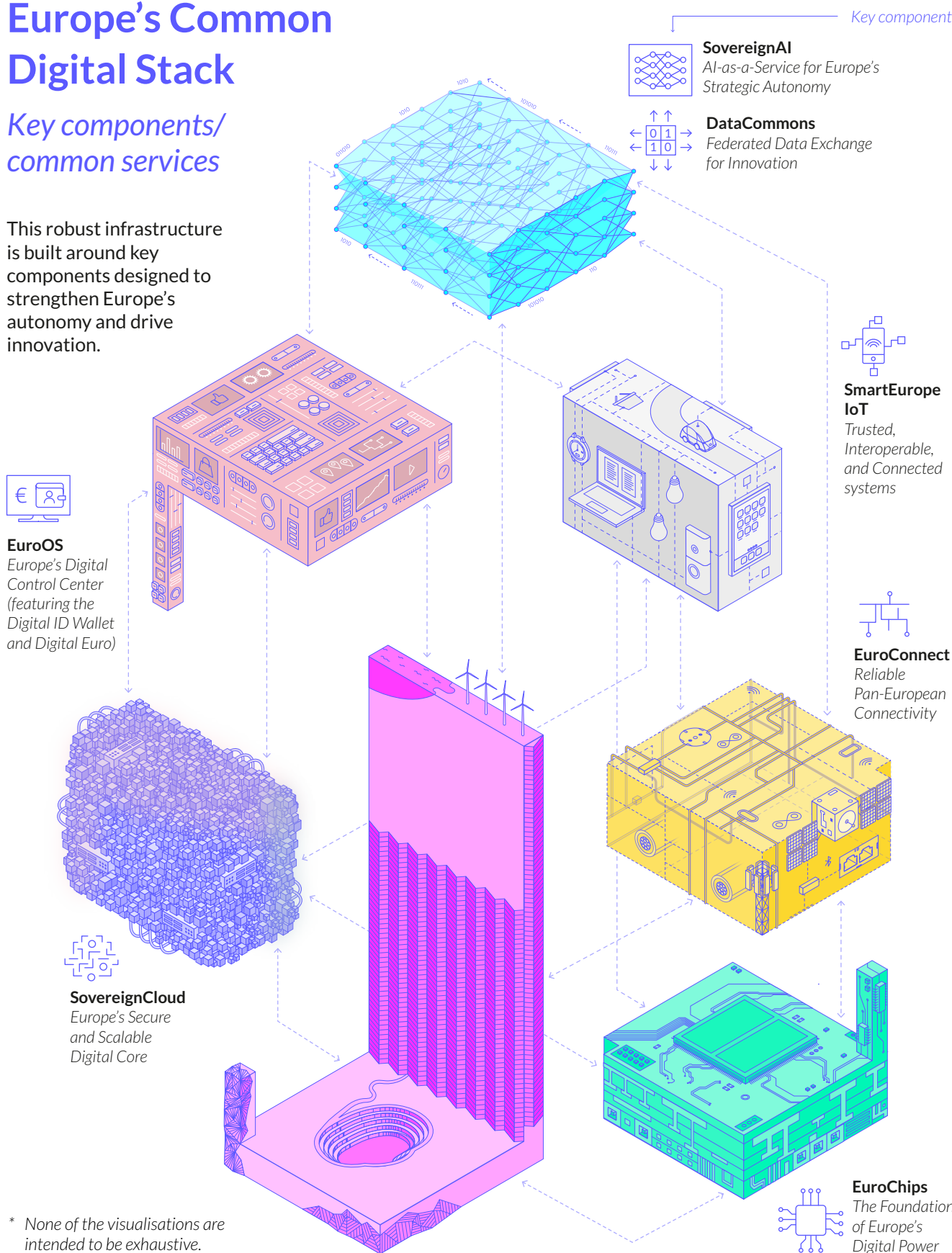
The layers



Europe's Common Digital Stack

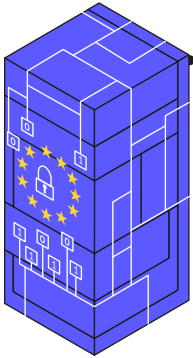
Key components/ common services

This robust infrastructure is built around key components designed to strengthen Europe's autonomy and drive innovation.



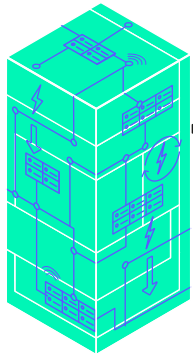
The EuroStack

Core principles



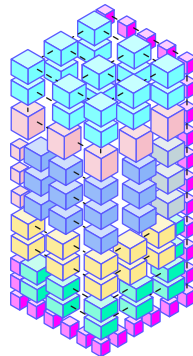
Sovereignty and security

Guaranteeing that Europe's critical digital infrastructure is under European jurisdiction, protected by robust security-by-design and privacy-by-design.



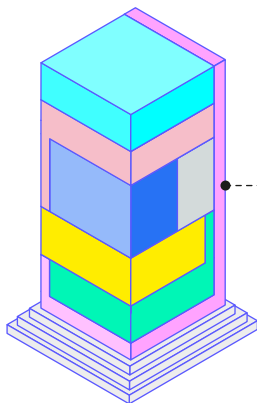
Sustainability

Building energy-efficient and resource-resilient systems to meet Europe's Environmental and Climate objectives.



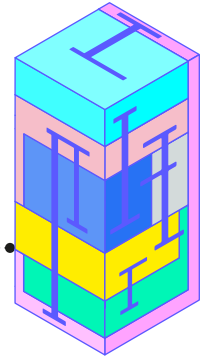
Decentralised sovereign infrastructure

Combining edge computing and centralized systems for improved efficiency and data sovereignty.



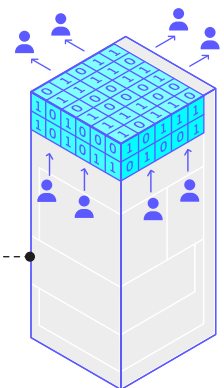
Strong democracy

Digital technologies that do no harm and fundamentally strengthen democratic societies.



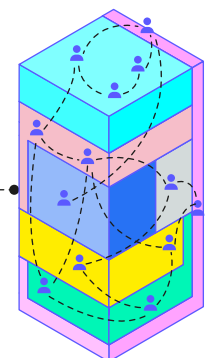
De-Proprietaryisation and interoperability

Promoting integration across an open source, federated tech stack, while reducing reliance on Big Tech proprietary solutions.



Data as common good

Treating data as a shared resource to unlock innovation while safeguarding societal interests and fundamental rights.



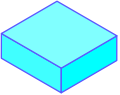
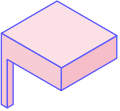
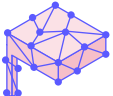
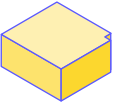
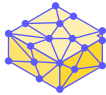
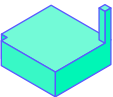
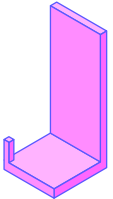
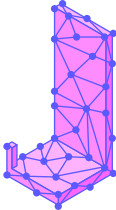
Inclusive governance

Ensuring harmonized regulations, accountability, and a balance between short-term resilience and long-term autonomy.

** None of the visualisations are intended to be exhaustive.*

The EuroStack

*Selection of Key companies,
Alliances and Networks per layer*

	KEY COMPANIES	ALLIANCES AND NETWORKS
Data and Artificial Intelligence 	- Mistral (France) - Aleph Alpha (Germany) - Siemens (Germany) - iGenius (Italy) - UiPath (Romania) - DeepL (Germany) - Celonis (Germany)	 - European AI Alliance - AI4EU Platform - Big Data Value Association (BDVA)
Software 	- SAP (Germany) - Nextcloud (Germany) - LibreOffice (Germany) - Dassault Systèmes (France) - Thales (France) - Bitdefender (Romania)	 - OpenForum Europe - European Software Strategy Alliance (ESSA) - Secure Identity Alliance
Internet of Things & Devices 	- Siemens (Germany) - Schneider Electric (France) - Bosch (Germany) - Philips (Netherlands) - Atos (France)	 IoT European Platforms Initiative (IoT-EPI)
Cloud 	- OVHcloud (France) - Scaleway (France) - Deutsche Telekom (Germany) - Schwarz (Germany)	 European Alliance for Industrial Data, Edge, and Cloud
Networks 	- Nokia (Finland) - Ericsson (Sweden) - Alcatel Submarine Networks (France) - Orange (France) - Airbus Defence and Space (Germany/France)	 5G Infrastructure Public Private Partnership (5G PPP) - Smart Networks and Services Joint Undertaking for 5G and 6G
Chips 	- ASML (Netherlands) - STMicroelectronics (France/Italy) - Infineon Technologies (Germany) - NXP Semiconductors (Netherlands) Key R&D Innovators: - Fraunhofer Society (Germany) - IMEC (Belgium)	 - Important Projects of Common European Interest (IPCEI) on Microelectronics - Silicon Saxony - European Semiconductor Industry Association (ESIA) - Industrial Alliance on Processors and Semiconductor Technologies
Raw Materials, Energy and Water 	- Umicore (Belgium) - Boliden (Sweden)	 European Raw Materials Alliance (ERMA)

** None of the visualisations
are intended to be exhaustive.*

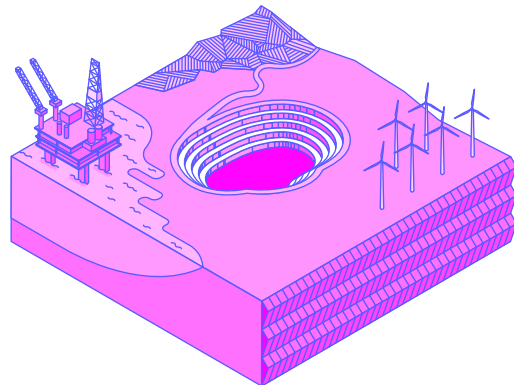
The separate layers explained

Raw Materials, Energy and Water

The Hidden Backbone of Technology

This foundational layer includes critical materials like rare earth elements, energy sources, and skilled workers, forming the bedrock for all digital infrastructure.

Key segments:
AI, data

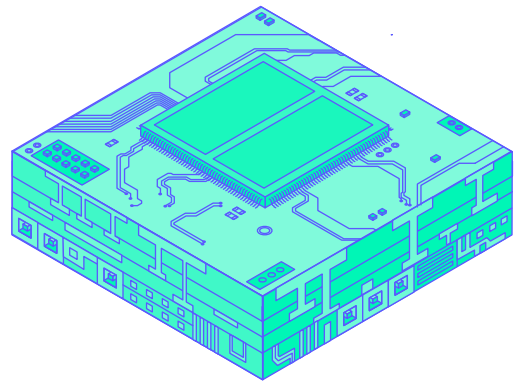


Chips

The Brain of the Stack

Processors and memory technologies, essential for powering digital infrastructure and ensuring secure supply chains, GPUs, and emerging quantum technology.

Key segments:
Foundry, design, equipment

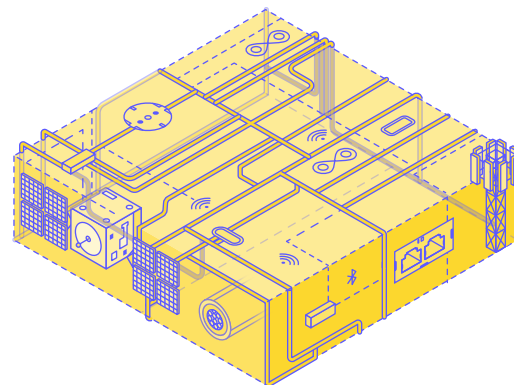


Networks

The Lifelines of Connectivity

Physical and digital connections, such as cell towers, fiber-optic networks, undersea cables, and the public core of the Internet, that link Europe to the global digital ecosystem.

Key segments:
TMC equipment, mobile networks, satellites, submarine communication cables

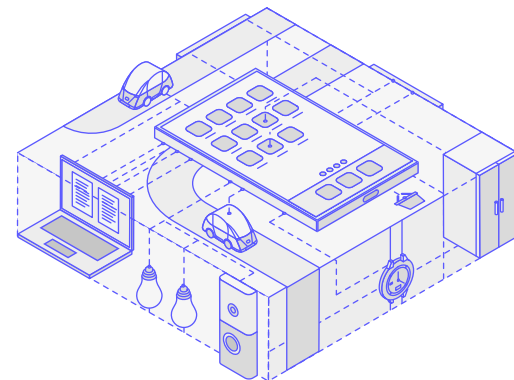


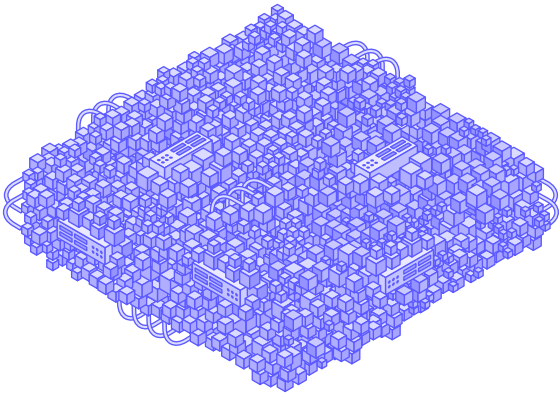
Internet of Things & Devices

The Connective Tissue of the Digital Ecosystem

From smartphones and laptops to IoT devices, enabling real-time information processing and data collection.

Key segments:
Smartphones, laptops, smart home devices, wearables, industrial & automotive IoT





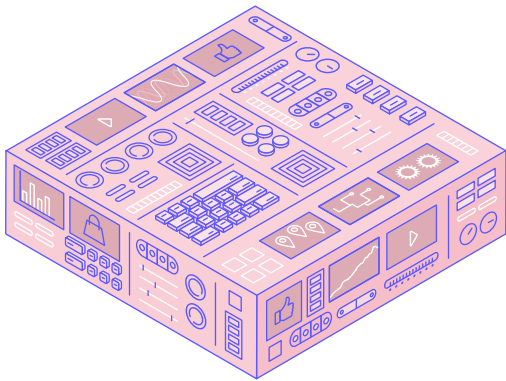
Cloud

The Operational Core of the Digital Economy

Secure data storage and computational power, distributed computing infrastructures which are central to data sovereignty and autonomy.

Key segments:

Data centres, cloud services



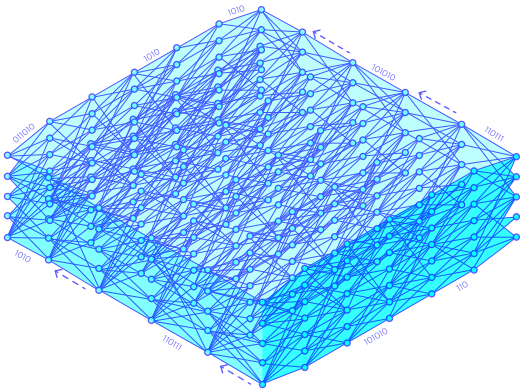
Software

The Command Centre of Platforms and Applications

Operating systems, applications, authentication and cybersecurity frameworks that drive digital interactions.

Key segments:

OS, online platforms, programs



Data and Artificial Intelligence

Catalysts for Innovation and Strategic Autonomy

Secure data storage and computational power, distributed computing infrastructures which are central to data sovereignty and autonomy.

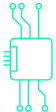
Key segments:

AI, data

* None of the visualisations are intended to be exhaustive.

The Geopolitical Dimension

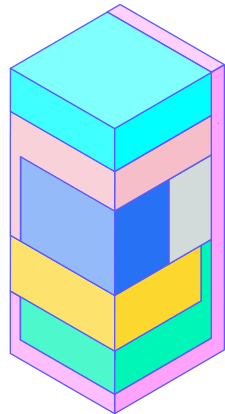
Most important countries of each stack layer and their key firms

KEY COUNTRIES					KEY FIRMS	
Data and Artificial Intelligence		 US		 China	OpenAI, Microsoft, Google, Meta, Anthropic, XAI, Amazon, Baidu, Tencent, Alibaba, DeepSeek	
Software		 US	 China	 Germany	Microsoft, Apple, Alphabet, Meta, Amazon, Salesforce, SAP, ByteDance, Tencent	
Cloud		 US		 China	Amazon, Microsoft, Alphabet, Alibaba	
Internet of Things & Devices		 US	 China	 Korea	 Germany	Amazon, Google, Apple, Samsung, Huawei, Bosch, Siemens, Xiaomi
Networks		 US	 China	 Europe	 Japan	Huawei, Nokia, Ericsson, ZTE, SpaceX, NEC
Chips		 Taiwan	 Korea	 US	 Netherlands	TSMC, Samsung, Intel, NVIDIA, AMD, ASML
Raw Materials, Energy and Water		 US	 China	 Russia		Chinese government (through SOEs e.g., China Rare Earth Group), ExxonMobil, Gazprom

The EuroStack

European Policies operating on the stack

1 Policies that apply to the whole stack



- Digital Decade
- A New European Innovation Agenda, European Innovation Ecosystems (EIE)



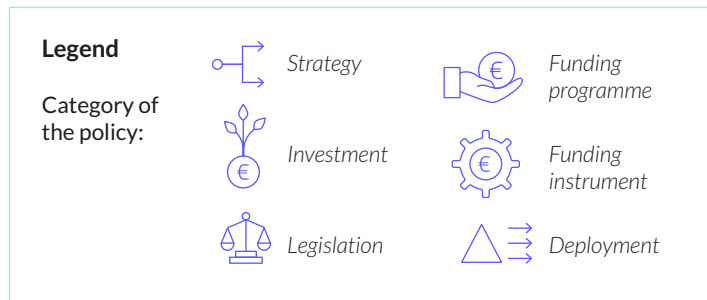
- InvestEU
- The Recovery and Resilience Facility (NextGenerationEU)
- VentureEU
- European Innovation Council (EIC) Fund
- European Tech Champions Initiative



- EU Cybersecurity Strategy, Cyber Security Act, Action Plan Cybersecurity and Health

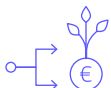
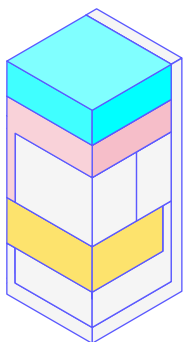


- Digital Europe Programme (DIGITAL)
- Horizon Europe



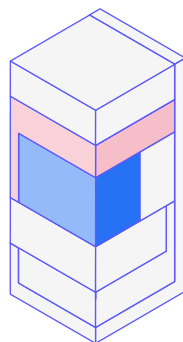
2 Policies that apply to multiple stack layers

Data and Artificial Intelligence, Software, Networks



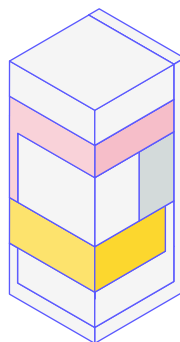
- The Common Security and Defence Policy (CSDP)
- European Defence Fund

Software, Cloud



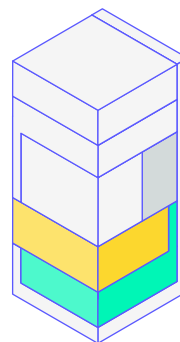
- Digital Services Act (DSA)
- Digital Markets Act (DMA)
- EU Cloud and AI Development Act

Software, Internet of Things & Devices, Networks,



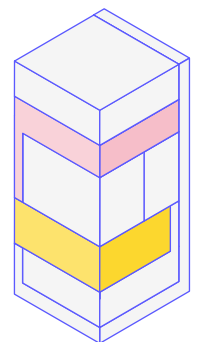
- Network and Information Systems Directive (NIS2 Directive)
- Digital Operational Resilience Act (DORA)

Internet of Things & Devices, Networks, Chips



- Quantum Technologies Flagship

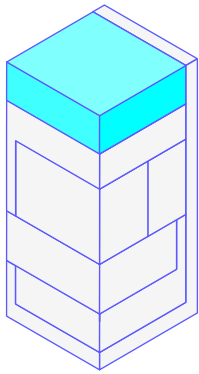
Software, Networks



- Connecting Europe Facility (CEF)

3 Policies that apply to certain layers of the stack

Data and Artificial Intelligence



- European AI strategy (2021 Review)
- EU Artificial Intelligence Act
- AI Innovation Package

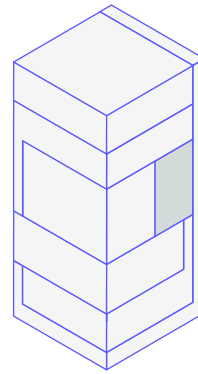


- Data Governance Act (DGA) + Data Act
- General Data Protection Regulation (GDPR).



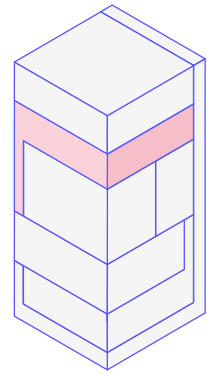
- European Strategy for Data (European Health Data Space)
- AI Factories Initiative
- Apply AI
- AI in Science
- Data Union Strategies

Internet of things & Devices

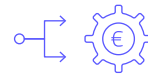


- Cyber Resilience Act

Software

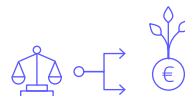
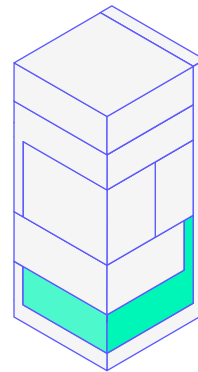


- eIDAS 2.0
- EU Digital Identity Wallet (EUDI)



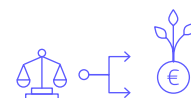
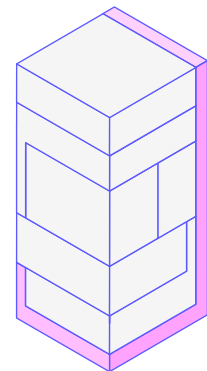
- The Common Foreign and Security Policy (CFSP)
- European Peace Facility (EPF)

Chips



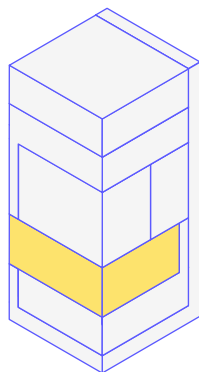
- Chips Act
- Chips for Europe Initiative

Raw Materials, Energy and Water



- Critical Raw Materials Act

Networks



- European Quantum Communication Infrastructure (EuroQCI)



- CASSINI (Space and Defence innovation)



- 5G Action Plan
- 5G PPP
- European Electronic Communications Code

* None of the visualisations are intended to be exhaustive.

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